

TARAWA BONRIKI, Kiribati

WMO: 916100

Lat: 1.382N

Lon: 173.147E

Elev: 3

StdP: 101.29

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	24.4	24.9	21.0	15.6	27.4	21.4	16.1	27.6	11.5	28.8	10.2	28.6	2.8	70	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	4.6	32.2	26.7	32.0	26.6	31.7	26.5	28.0	30.8	27.6	30.6	27.3	30.4	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	22.9	30.1	26.7	22.3	29.8	26.4	21.9	29.5	89.4	30.9	87.8	30.7	86.7	30.9	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.2	9.1	8.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB	21.2	29.1	1.6	1.2	20.0	30.0	19.1	30.6	18.2	31.3	17.0	32.1
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	28.4	28.4	28.2	28.4	28.6	28.7	28.6	28.7	28.9	28.8	28.8	28.6	(6)
(7)		DBStd	0.87	0.86	0.94	0.91	0.85	0.79	0.81	0.79	0.80	0.82	0.79	0.81	0.96	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6782	569	514	565	552	578	561	578	579	566	582	563	575	(10)
(11)		CDD18.3	3741	311	281	307	302	319	311	319	321	316	324	313	317	(11)
(12)		CDH23.3	44441	3586	3209	3489	3495	3824	3742	3897	3897	3869	3923	3776	3734	(12)
(13)	CDH26.7	16194	1209	1058	1116	1179	1406	1416	1495	1490	1523	1516	1443	1342	(13)	
(14)	Wind	WSAvg	4.2	5.1	5.3	4.8	4.3	4.0	3.6	3.7	4.3	3.9	4.2	4.1	(14)	
(15)	Precipitation	PrecAvg	1708	191	146	194	184	127	128	112	118	95	100	113	201	(15)
(16)		PrecMax	3275	511	449	538	518	372	324	255	364	315	393	372	513	(16)
(17)		PrecMin	512	30	8	23	18	35	31	33	10	8	6	10	11	(17)
(18)		PrecStd	837	143	111	150	138	87	83	61	84	77	101	88	152	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	32.2	32.1	31.7	32.1	32.2	32.2	32.2	32.2	32.3	32.2	(19)	
MCWB			26.9	26.8	27.0	26.8	26.8	26.9	26.8	26.5	26.4	26.6	26.5	26.9	(20)	
2%		DB	31.3	31.6	31.2	31.2	31.6	31.7	31.7	31.8	31.9	31.9	31.9	31.8	(21)	
		MCWB	26.5	26.5	26.7	26.7	26.7	26.6	26.7	26.4	26.3	26.4	26.4	26.8	(22)	
5%		DB	30.9	31.0	30.7	30.9	31.1	31.2	31.2	31.3	31.4	31.4	31.5	31.2	(23)	
		MCWB	26.4	26.3	26.5	26.6	26.6	26.5	26.6	26.3	26.2	26.2	26.3	26.6	(24)	
10%		DB	30.4	30.3	30.2	30.4	30.8	30.9	30.9	31.0	31.0	31.0	31.1	30.8	(25)	
		MCWB	26.1	26.1	26.3	26.4	26.5	26.4	26.6	26.2	26.2	26.0	26.2	26.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	27.9	28.0	28.0	28.0	27.8	27.8	27.9	27.7	27.9	28.1	(27)	
MCDWB			30.6	30.7	30.6	30.5	30.7	30.7	30.8	30.6	30.8	31.0	31.2	30.9	(28)	
2%		WB	27.4	27.3	27.4	27.3	27.4	27.4	27.3	27.2	27.2	27.2	27.3	27.6	(29)	
		MCDWB	30.2	30.1	30.1	30.1	30.3	30.4	30.6	30.6	30.7	30.9	30.8	30.6	(30)	
5%		WB	27.0	26.9	27.0	27.0	27.1	27.1	27.0	26.8	26.9	26.7	26.9	27.1	(31)	
		MCDWB	30.0	29.7	29.8	29.9	30.1	30.2	30.5	30.3	30.5	30.4	30.5	30.2	(32)	
10%		WB	26.6	26.6	26.7	26.7	26.8	26.7	26.7	26.5	26.6	26.4	26.6	26.7	(33)	
		MCDWB	29.5	29.4	29.4	29.5	29.8	29.8	30.0	30.0	30.1	30.0	30.0	29.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.3	4.2	4.3	4.2	4.4	4.7	4.7	4.6	4.8	4.7	4.5	(35)	
MCDBR			4.6	4.6	4.7	4.5	4.8	5.0	5.0	5.0	5.2	5.1	4.9	(36)		
MCWBR			2.6	2.4	2.5	2.2	2.3	2.4	2.5	2.4	2.4	2.5	2.5	2.4	(37)	
5% WB		MCDBR	4.5	4.3	4.4	4.3	4.6	4.9	4.8	4.8	4.8	5.0	4.9	4.7	(38)	
		MCWBR	2.8	2.6	2.7	2.3	2.4	2.5	2.5	2.5	2.6	2.6	2.5	2.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.423	0.424	0.424	0.403	0.389	0.384	0.384	0.380	0.384	0.379	0.389	0.416	(40)	
taud		2.499	2.487	2.514	2.586	2.627	2.640	2.642	2.646	2.617	2.631	2.602	2.524	(41)		
Ebn at Noon		906	908	902	901	892	885	889	910	924	939	932	907	(42)		
Edh at Noon		112	115	112	101	94	91	92	94	99	98	100	108	(43)		
(44)	All-Sky Solar Radiation	RadAvg	5.53	5.92	5.84	5.95	5.94	5.70	5.74	6.13	6.45	6.46	6.15	5.48	(44)	
(45)		RadStd	0.84	0.77	0.91	0.61	0.39	0.44	0.40	0.47	0.47	0.52	0.43	0.81	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page